

Model Fairness Audit: Loan Approval v2 (synthetic)

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1. Executive summary

69 /100
Fairness
score

Verdict: Needs attention

Primary metric: Equal opportunity. Are truly qualified people approved at similar rates?

Pass threshold: True-positive-rate ratio of at least 0.80.

Key findings

- 1 of 11 groups fail equal opportunity; the most affected is 'Female, 45plus' (value 0.667).
- For context, the lowest selection-rate ratio is 0.614 (the four-fifths rule requires 0.80).
- Overall drift is significant (highest PSI 0.6844); 1 feature(s) shifted.
- Decision explanations reproduce the model's decisions on 85% of records.

Priority actions

- **High** Investigate the equal opportunity gap for 'Female, 45plus' (gender x age_band): measured 0.667; required: true-positive-rate ratio of at least 0.80.
- **Medium** Retrain with recent data: 'income' has drifted (PSI 0.6844, significant).
- **Low** Schedule a fairness and drift re-check every quarter.

2. Scope: data and model

Model	Loan Approval v2 (synthetic)
Intended use	Lending
Records audited	4,000
Columns	7
Model decision column	predicted_approval
Actual outcome column	loan_repaid
Sensitive attributes	gender, age_band
Overall selection rate	74%
Overall accuracy	80%
Positive outcome base rate	84%

3. Method and metric selection

Records are grouped by each sensitive attribute and by every combination of two or more attributes (intersectional groups). Each group is compared with a reference group: the best-off group with at least 100 records. The primary metric, **Equal opportunity**, decides pass or fail; the other metrics are reported for context.

Metric	Question answered	Pass threshold
Demographic parity	Are groups selected at similar rates?	Selection-rate ratio of at least 0.80
Equal opportunity	Are truly qualified people approved at similar rates?	TPR ratio of at least 0.80
Equalized odds	Are misses and false alarms similar across groups?	TPR and FPR gaps of at most 0.10
Predictive parity	Does a positive decision mean the same for every group?	Precision ratio of at least 0.80

Groups below 100 records are marked low confidence; if they fail they are marked "Review" rather than "Fail". Selection rates include 95% Wilson confidence intervals. The fairness score blends average and worst-group compliance with the primary metric, minus a penalty per failing group, and is capped below 70 whenever any group fails.

4. Fairness results: single attributes

gender

Group	Size	Selection rate (95% CI)	Selection ratio	TPR	FPR	TPR ratio	Odds gap	Precision ratio	Equal opportunity	Status
Male (reference)	2,215	81% (80%-83%)	1.00	89%	45%	1.00	0.00	0.96	1.00	Pass
Female	1,785	66% (64%-68%)	0.81	75%	21%	0.84	0.23	1.00	0.84	Pass

age_band

Group	Size	Selection rate (95% CI)	Selection ratio	TPR	FPR	TPR ratio	Odds gap	Precision ratio	Equal opportunity	Status
25-44 (reference)	2,027	79% (77%-81%)	1.00	87%	40%	1.00	0.00	0.97	1.00	Pass
under25	817	78% (75%-81%)	0.99	86%	37%	0.99	0.04	0.98	0.99	Pass
45plus	1,156	64% (61%-66%)	0.81	72%	21%	0.83	0.19	1.00	0.83	Pass

5. Fairness results: intersectional groups

gender x age_band

Group	Size	Selection rate (95% CI)	Selection ratio	TPR	FPR	TPR ratio	Odds gap	Precision ratio	Equal opportunity	Status
Male, 25-44 (reference)	1,116	86% (83%-88%)	1.00	92%	52%	1.00	0.00	0.93	1.00	Pass
Male, under25	455	84% (80%-87%)	0.98	90%	47%	0.98	0.04	0.94	0.98	Pass
Male, 45plus	644	73% (69%-76%)	0.85	81%	30%	0.88	0.21	0.96	0.88	Pass
Female, under25	362	71% (66%-76%)	0.83	80%	23%	0.87	0.28	0.98	0.87	Pass
Female, 25-44	911	71% (68%-74%)	0.83	80%	27%	0.87	0.25	0.97	0.87	Pass
Female, 45plus	512	53% (48%-57%)	0.61	61%	10%	0.67	0.41	1.00	0.67	Fail

6. Drift analysis

Population Stability Index (PSI): below 0.10 stable, 0.10 to 0.25 moderate, above 0.25 significant. The Kolmogorov-Smirnov p-value is given for numeric features (below 0.05 indicates a significant shift).

Feature	PSI	KS p-value	Band
income	0.684	0.0	Significant
loan_amount	0.086	0.0	Stable
employment_years	0.000	1.0	Stable

7. Explainability

Explanations are produced by a transparent surrogate model that reproduces the audited model's decisions on **85%** of records. Sensitive attributes are excluded from explanations and counterfactuals.

Overall drivers of the model's decisions

Feature	Share of influence
income	42%
employment years	37%
loan amount	21%

Example decisions

Record #0 **Rejected**

The model rejected this record, mainly because of income (26,091) and employment years (3), compared with a typical record.

Main factors: income = 26,091 (hurt), employment years = 3 (hurt), loan amount = 17,262 (hurt)

What would change the outcome

- If income were increased from 26,091 to 46,432, the decision would likely be approved.
- If employment years were increased from 3 to 15, the decision would likely be approved.

Record #1 Approved

The model approved this record, mainly because of loan amount (6,851) and income (50,880), compared with a typical record. Employment years (6) counted against it, but not enough to change the outcome.

Main factors: loan amount = 6,851 (helped), employment years = 6 (hurt), income = 50,880 (helped)

What would change the outcome

- If income were reduced from 50,880 to 27,149, the decision would likely be rejected.
- If loan amount were increased from 6,851 to 27,144, the decision would likely be rejected.

Possible proxy features

No feature is strongly associated (Cramer's V of 0.5 or more) with a sensitive attribute.

8. Limitations

- Fairness metrics can conflict mathematically when groups have different base rates; passing the primary metric does not imply passing every metric.
- Results describe the supplied data only. If outcomes were recorded under a biased historical process, outcome-based metrics inherit that bias.
- Groups with fewer than 100 records are reported with low confidence and are not counted as firm failures.
- Per-decision explanations come from a transparent surrogate model that approximates the audited model; its agreement rate is stated in section 6.
- Counterfactuals change one feature at a time within the observed range and indicate sensitivity; they are not guaranteed routes to a different outcome.
- Proxy detection measures statistical association only and does not show that the model relies on the feature.
- This report supports, but does not replace, legal and domain review.

9. Recommendations

- High Investigate the equal opportunity gap for 'Female, 45plus' (gender x age_band): measured 0.667; required: true-positive-rate ratio of at least 0.80.
- Medium Retrain with recent data: 'income' has drifted (PSI 0.6844, significant).
- Low Schedule a fairness and drift re-check every quarter.

10. Audit trail and sign-off

Report generated	2026-10-07 12:14 UTC
Tool	FairLens 1.2 (Python 3.11.15)
Groups tested	11 (0 with fewer than 100 records)

Role	Name	Signature	Date
Model owner			
Independent reviewer			
Compliance / risk			